Ankit Shukla

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Senior Software Engineer | C/C++ | Linux Systems | GPU & Linux Graphics Stack | Automotive systems| Embedded & Avionics Systems | Cloud Native

PROFESSIONAL SUMMARY

Senior Software Engineer with 14+ years building and shipping systems-level C/C++ software across Linux, Windows, QNX, and WinCE. Deep expertise in GPU driver development, OpenGL, and the graphics rendering stack (DRM, LLVM-based GL pipeline), multithreaded and distributed systems, Linux internals, IPC, and socket programming, spanning storage, automotive, avionics, and graphics domains. Most recently, designed and built microservices-based, cloud-native components exposing REST and gRPC APIs and backed by SQL and NoSQL data stores for resource-management workloads on Kubernetes. Delivered DO-178B safety-critical avionics software communicating over ARINC 429 and ARINC 664 (AFDX) networks, and CAN bus-integrated automotive infotainment and instrument cluster systems. Proven track record designing resource-management and licensing systems, delivering cloud-integrated services on AWS and Azure, and taking features from design through implementation, testing, and production on-call support. Experienced technical leader who has run a team of six, mentored engineers through code review and design discussions, and worked comfortably from device-driver, kernel-adjacent, and GPU-adjacent code up through cloud services.

PROFESSIONAL HIGHLIGHTS

- 14+ years in Systems Software Development
- Microservices Architecture & APIs — gRPC and REST API design for cloud-native resource-management services
- SQL & NoSQL Databases — data modeling and integration for distributed backend services
- GPU Driver Development & OpenGL — DRM subsystem, graphics pipeline, LLVM-based GL stack
- Expert in Modern C++ (11/14/17)
- DO-178B Avionics Software — ARINC 429 & ARINC 664 (AFDX) protocols
- Containerized Applications & Kubernetes — Docker, orchestration, cloud-native deployment
- Linux Systems Programming
- Embedded Linux & RTOS (QNX, WinCE, LynxOS)
- Automotive Bus Systems — CAN
- Multithreading & Concurrency / Distributed Systems
- Technical Leadership (led team of 6)
- Cross-platform Development
- Production Debugging & On-call Support
- Cloud-Integrated Infrastructure (AWS, Azure)

TECHNICAL SKILLS

Programming Languages: C, C++, Python, Go, Java

Modern C++: STL, Templates, Smart Pointers, RAII, Move Semantics, Lambda Expressions, Design Patterns (Singleton, Factory, Adapter, Builder), SOLID

Architecture & APIs: Microservices Architecture, gRPC, REST API Design, JSON/XML, SOAP, XML-RPC

Databases: SQL Databases, NoSQL Databases, Data Modeling for Distributed Services

Graphics & GPU: OpenGL, GPU Driver Development, DRM (Direct Rendering Manager) Subsystem, LLVM-based Graphics Pipeline, Hardware-Accelerated Rendering, Radeon GPU Architecture

Systems Programming: Linux, POSIX, IPC, Socket Programming, TCP/IP, Memory Management, Multithreading, Synchronization, Performance Optimization

Embedded Systems: Embedded Linux, QNX, WinCE, LynxOS, ARM, x86, Kernel-adjacent Development, Device Drivers

Avionics: DO-178B Software Development Lifecycle, ARINC 429, ARINC 664 (AFDX), XML-RPC

Automotive: CAN Bus, Vehicle Network Signal Processing, Infotainment & Instrument Cluster Systems

Distributed Systems & Cloud: AWS, Azure, REST, gRPC, SOAP, JSON/XML, OpenSSL 3.0

Containers & Orchestration: Docker, Kubernetes, Containerized Application Deployment, Cloud-Native Architecture

Testing: GoogleTest, GoogleMock, VectorCAST, Ginkgo, TestRail
Security & Compliance: SBOM (Software Bill of Materials), JFrog Artifactory, Black Duck
Build Tools: CMake, Make, Bazel, Apache Maven, MS Build, Jenkins
Debugging: GDB, Trace32, Lauterbach, Core Dump Analysis, Root Cause Analysis
Version Control & PM: Git, SVN, RTC, Bitbucket, Gerrit, Jira, IBM Rational DOORS
AI-assisted Development: Cursor, GitHub Copilot with Claude models

PROFESSIONAL EXPERIENCE

Software Engineer – Resource Manager (Heimdall) | Storage | Cohesity Inc.
Dec 2024 – Present

Built core components of Heimdall, an internal resource manager that provisions and tracks Linux/Windows VM fleets, disks, and containerized workloads used by backup services, in C++, Python, and Go.

- Designed and implemented Heimdall as a microservices-based system, with services communicating over gRPC and exposing REST APIs consumed by dependent teams and internal tooling.
- Modeled and integrated both SQL and NoSQL databases for tracking VM fleet, disk, and workload state, choosing the storage approach based on each service's consistency and query needs.
- Worked with Kubernetes-orchestrated, containerized services as part of Heimdall's cloud-native deployment model, coordinating provisioning across containers alongside traditional VM fleets.
- Owned features end-to-end: wrote design docs, drove requirement discussions with dependent teams, broke work into epics/stories in Jira, and implemented and code-reviewed the resulting changes.
- Wrote unit tests (GoogleTest/GoogleMock) and integrated automated test runs through Ginkgo, publishing results to TestRail for QA sign-off.
- Investigated and resolved release-blocking defects, including issues surfaced during on-call rotations for customer-impacting backup failures.
- Maintained the build/image scripts used to produce Heimdall deployment images across environments, including generating SBOMs (Software Bill of Materials) for build artifacts to support vulnerability tracking and compliance requirements.

Software Engineer – Shared Licensing Component (SLIC) | Storage | Veritas Technologies
Jan 2023 – Dec 2024

- Developed licensing/entitlement features in C++ for SLIC, the shared license-validation component used across Veritas's product line, built on OpenSSL and Poco.
- Designed and implemented on Windows, then ported and validated changes across Red Hat, OpenSUSE, AIX, and Solaris.
- Rebuilt the Poco and OpenSSL dependencies on each supported platform and delivered updated SLIC SDKs to downstream product teams to close known CVEs.
- Triaged vulnerabilities flagged by Black Duck in OpenSSL/Poco and assessed real-world impact on SLIC before prioritizing fixes.
- Published and maintained SLIC build artifacts in JFrog Artifactory and generated SBOMs (Software Bill of Materials) for the component to support downstream CVE tracking and compliance sign-off.

Senior Software Engineer / Technical Lead – Radio Block Center (RBC) | Rail / Mobility | Siemens Mobility
Mar 2019 – Jan 2023

- Led a team of 6 engineers on RBC, safety-critical software that receives train position reports over Balise devices and issues movement authority to trains.
- Delivered new features in C/C++/Python from requirements through implementation, static analysis, and unit testing with VectorCast.
- Ran code reviews and diagnosed field issues from logs, driving root-cause analysis and fixes for the team.
- Built two internal automation tools in Python (Tkinter) — one to automate environment/software setup, another to update code baselines and timestamps across unit-test suites — both adopted team-wide to cut manual setup time.
- Owned monthly billing releases and forecast updates for the engagement.

Software Engineer – Automotive Infotainment & Instrument Cluster (QNX) | Automotive | Tata Elxsi Ltd
Jun 2015 – Mar 2019

- Applink (Panasonic Automotive, 3 yrs 2 mo): built C/C++ modules for a smartphone-to-infotainment connectivity system on QNX/Windows, covering voice command handling, touch UI input, and embedded navigation,

communicating over the vehicle's CAN bus; authored design docs, analyzed core dumps, and fixed defects found in production logs.
- GM Instrument Cluster (Visteon, 6 mo): processed CAN bus vehicle data and drove UI widgets showing tire pressure, fuel level, and speed/tachometer readings on QNX/Windows; delivered features from system design through GoogleTest/GoogleMock unit testing and static analysis.
- Used Singleton and related design patterns to keep shared hardware-access code consistent across both projects.

Software Engineer – Graphics Driver Development (AMD, WinCE) | Graphics / Multimedia | Infosys Ltd
Aug 2014 – Jun 2015

- Ported the Linux OpenGL/LLVM call stack to WinCE and worked on the Radeon GPU driver's DRM subsystem to bring hardware-accelerated rendering to WinCE-based boards (Ontario, Kabini, Mullins), building hands-on understanding of the full GPU driver and graphics rendering pipeline.
- Diagnosed and fixed GPU driver defects affecting video and gaming rendering performance, and built OpenGL test applications to validate rendering correctness across the graphics stack.

Software Engineer – Onboard Maintenance System (RTOS LynxOS) | Avionics | Rockwell Collins India Pvt. Ltd
Aug 2011 – Aug 2014

- Developed DO-178B compliant features in C/C++ for OMS, an avionics subsystem (Diagnostic Report and Display Manager applications) that communicates with aircraft LRUs over ARINC 429, ARINC 664 (AFDX)/Ethernet, and XML-RPC to surface maintenance data to flight crews.
- Worked on a 20-person program using Trace32/Lauterbach for hardware-level debugging on LynxOS targets, with requirements tracked in IBM Rational DOORS and verification evidence maintained per DO-178B objectives.
- Wrote and executed functional and unit tests and fixed defects found during integration testing.

EDUCATION

B.Tech — UPTU, 2010
Diploma in System Software — Pune University, 2011